

Physical Acoustics: Improving the Reverberation Time in the Student Success Building at UCA

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Executive Summary:

On April 10th, 2024, we assessed the physical acoustics of the University of Central Arkansas Student Success Center: specifically, the coaching room and commons area. A series of experiments were conducted to obtain the reverberation time in each room, including adding damping tiles and investigating their effects. Based on the data, recommendations were made to improve the speech intelligibility in each space based on the ANSI Standard S12.60 for Classroom Acoustics for furnished, unoccupied rooms. For the coaching room, 352-ft² of damping material can be added to bring the reverberation time down to the recommended 0.6-s. For the commons area, 448-ft² of damping material can be added to reduce the reverberation time to the recommended 0.7-s. One more important note is that these recommendations were made based on the assumption that the coaching door in the student success center remains closed.

Introduction:

For our final project for physical acoustics, we were tasked with conducting a series of experiments in a space on campus, analyzing the data, and making some recommendations to try to improve the acoustics of the space.

The UCA Student Success Center is used to help students achieve their greatest potential during their time on campus through coaching and tutoring. However, when multiple conversations are happening in the coaching room, communication is impaired due to the poor physical acoustics. The room had painted sheetrock walls and the only absorbing surface was the carpet on the floor. The commons area suffered from a similar problem, with the walls being made of the same material, the floor being partially covered, and the room being very large.

We assessed the acoustics of each room by popping balloons and recording the sound on two laptops using Audacity. We then saved it as a .wav file and used code based on Dan Russell from Penn State. We included a floor plan of the building below in Figure 1; the red outlined room is the coaching room and the blue outlined room in the main commons area.

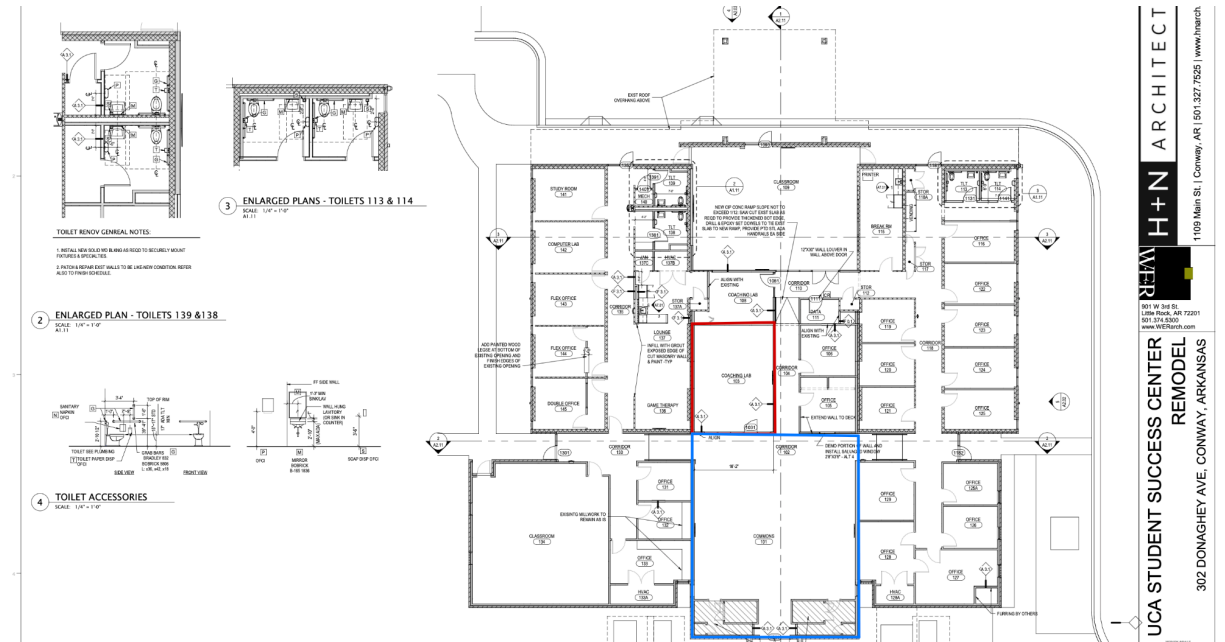


Figure 1. This is a floor plan of the student success building. The spaces analyzed are color coded. The red outline is the coaching room, and the blue outline is the commons area.

In the coaching room, we popped three balloons with the door open and three with the door closed to get some baseline readings. Then, we added some 2-ft by 2-ft ceiling tiles with an NRC value of 0.9 totaling to 96-ft² of total area. We popped three balloons again with the door open and then with the door closed while recording the data in each case. Finally, we added some additional 2-ft by 3-ft ceiling tiles with an NRC rating of 0.55 totaling to 80-ft² of total surface area, and we popped the balloons one last time. We also repeated this process in the larger commons area.

The NRC rating is useful, because it is a metric of how effective a material is at reducing reverberation time [1]. For example, a material with an NRC rating of 0.9 is twice as effective at reducing reverberation as a material with an NRC rating of 0.45 [1]. When there is loud sound in a closed space, the sound bounces off the walls and it takes time to reduce to nothing. The time it takes for the sound to dissipate is the reverberation time, and it follows an exponential decay curve. More specifically, it is the time it takes for the sound to decay to background noise or to have a 60-dB decrease from the maximum [1].

$$T \sim \frac{V}{A}, \quad (1)$$

where V is the volume of the space, A is the absorptive surface area, and T is the reverberation time. The reverberation time is directly proportional to the volume of the space and inversely proportional to the absorptive surface area of the space. The Student Success Center has poor acoustics because both areas have large volumes and a small amount of absorptive surface area. This results in larger reverberation times, which impair speech communication due to the sound echo reaching the ears at the same time the sound from the source arrives. The ANSI Standard

S12.60 for Classroom Acoustics states that the reverberation time be in the 0.6-0.7 second range for furnished but unoccupied spaces [1].

In music, sounds are grouped into octaves. An octave on a piano is the distance between one note and another with double the frequency. An octave band is the set of frequencies that fall within an octave [1]. To calculate the octave band range of frequencies, we can start with a center frequency f_c . To calculate the lower and upper limits of the octave, we can use

$$f_c = \sqrt{2}f_{\min} = \frac{f_{\max}}{\sqrt{2}} \quad (2)$$

In Equation 2, $f_{\min}$ is the lower frequency boundary and $f_{\max}$ is the upper frequency boundary. For acoustic measurements, its common to pick the center frequencies to be 125-Hz, 250-Hz, 500-Hz, 1000-Hz, 2000-Hz, and 4000-Hz. This frequency range covers the relevant frequencies that impact human hearing and speech intelligibility [1].

Each room is like a musical instrument; it has natural frequencies called acoustic modes. These modes depend on the dimensions of the room i.e. width, length, and height. The formula for calculate these room modes is

$$f_n = \frac{nv_s}{2L}, \quad (3)$$

where n is the n-th mode, v_s is the speed of sound, and L is the dimension being calculated [1].

On the day of the experiment, the temperature in the room was 68-F (20-C). At that temperature, the speed of sound in air is 343.5 meters per second. For the coaching room, the length is 292-in (44.648-m), the width is 237-in (6.0198-m), and the height at the tallest point is 211-in (5.3594-m). The ceiling slopes down to 123.5-in (3.1369-m) on one side. For the commons area, the height is 211-in (5.3594-m) in the center of the room. The ceiling slopes down to 123.5-in (3.1369-m) on both sides. The length is 454-in (11.5316-m) and the width is 440-in (11.176-m). Table 1 below shows the four lowest modes calculated using Equation 3 above. The height mode was calculated using the tallest height in the room. Long reverberation times in the lowest octave band are most likely associated with the room height.

	dimensions	1st Mode (Hz)	2nd Mode (Hz)	3rd Mode (Hz)	4th Mode (Hz)
Coaching Room	Length:	28.5308482	57.0616964	85.5925446	114.1233928
Coaching Room	Width:	23.15688707	46.31377413	69.4706612	92.62754827
Coaching Room	Height:	32.04649774	64.09299548	96.13949323	128.185991
Commons Area	Length:	15.36775233	30.73550465	46.10325698	61.47100931
Commons Area	Width:	14.89385688	29.78771376	44.68157064	59.57542752
Commons Area	Height:	32.04649774	64.09299548	96.13949323	128.185991

Table 1. These are the lowest frequency room acoustic modes for the coaching room and common area room of the Student Success Center.

Results/Analysis:

Part 1: Experimental Setup

Figures 2 and 3 below show the experiment setup in each room of the Student Success Center. In each case, we acquired data from two laptops located in different areas of the room. Balloons were inflated with an air pump and then popped with a bobby pin as close to the center of the room as possible. Then, the Linux application Audacity was used to capture the decaying sound and save it as a .wav file for future analysis.



Figure 2. This is the experimental setup in the coaching room of the student success building.



Figure 3. This shows the experimental setup in the commons area of the student success building.

Part 2: Experimental Data and Recommendations

For a space of 10,000 cubic feet or less, the ANSI Standard S12.60 recommends that the reverberation time is 0.6-seconds or less for furnished, unoccupied classrooms. For spaces between 10,000 cubic feet and 20,000 cubic feet, the recommendation is 0.7-seconds.

If we convert those to cubic meters, a reverberation time of 0.6-seconds is recommended for spaces 283 cubic meters or less, and a reverberation time of 0.7-seconds is recommended for spaces between 283 cubic meters and 566 cubic meters. Therefore, since the coaching room in the Student Success Center is approximately 190 cubic meters, the recommended reverberation time is 0.6-seconds or less for that room. Contrastingly, the commons area is approximately 547 cubic meters, so the reverberation time for that space should ideally be no larger than 0.7-seconds.

In the introduction, we stated that three balloons were popped for each scenario. Once we had collected that data, we obtained the average reverberation time for each octave band by calculating the average of all three instances. Then, we used the Python Acoustics library to

process the .wav files obtained using Audacity (see Appendix B). Tables 2 and 3 below show the experimental data for the coaching room with the door shut (and open) for all three scenarios: no damping, damping with the smaller panels, and damping with all panels. The T_{60} reverberation time in seconds is given for the octave bands discussed previously along with the average T_{60} value for each laptop and the total average T_{60} time. To specify further, the total average T_{60} time was obtained by averaging the average T_{60} value from each laptop. Finally it is important to note that the smaller 2-ft by 2-ft damping tiles have an NRC rating of 0.9, whereas the larger 2-ft by 3-ft damping tiles have an NRC rating of 0.55.

With no damping, the room's average T_{60} value is better with the door open at 1.09-s versus 1.22-s with the door shut. However, neither case is ideal. As we added damping tiles, the average T_{60} value worsened with the door open. For this reason, it is recommended to leave the door shut. With the doors shut, we were able to reduce the average T_{60} time by 0.27-s. This improvement should scale linearly, so we could add two times the amount of damping tiles to bring the average T_{60} value down to 0.68-s, which would be in the ideal range. Since 176-ft² of damping tiles were used in total, we could double that to 352-ft² of damping tile to have the ideal resonance. Additionally, the room has enough available area on the walls to handle the square footage of damping tiles needed.

Centre of Study Room - Door Shut - No Damping			Centre of Study Room - Door Shut - Damping Panels (Small)			Centre of Study Room - Door Shut - Big and Small Damping Panels		
Octave Bands	t60 Value (Simon)	t60 value (Slaton)	Octave Bands	t60 Value (Simon)	t60 value (Slaton)	Octave Bands	t60 Value (Simon)	t60 value (Slaton)
125	1.71130898	2.8668416	125	1.15480871	2.86583113	125	1.17455615	2.52721362
250	1.01748808	1.12786868	250	0.93562006	0.95129305	250	0.81835836	0.85171008
500	1.00346912	0.97952609	500	0.92788621	0.80688941	500	0.78894915	0.68133393
1000	1.04215376	0.99634892	1000	0.90365529	0.80993573	1000	0.7916897	0.71423429
2000	1.02592178	0.99922525	2000	0.99409056	0.83490081	2000	0.83467639	0.72393412
4000	0.95033505	0.92425105	4000	0.86311178	0.77883198	4000	0.83949193	0.71504801
Average	1.125112795	1.315676932	Average	0.963195435	1.174613685	Average	0.87462028	1.035579008
Total Average		1.220394863	Total Average		1.06890456	Total Average		0.9550996442

Table 2. This shows the experimental data for the study room with the door shut with no damping, damping with the 2-ft by 2-ft panels, and damping with the 2-ft by 2-ft panels combined with the 2-ft by 3-ft panels. The T_{60} reverberation time in seconds is given for the octave bands discussed previously along with the average T_{60} value for each laptop and the total average T_{60} time.

Centre of Study Room - Door Open - No Damping			Centre of Study Room - Door Open - Damping Panels (Small)			Centre of Study Room - Door Open - Big and Small Damping Panels		
Octave Bands	t60 Value (Simon)	t60 value (Slaton)	Octave Bands	t60 Value (Simon)	t60 value (Slaton)	Octave Bands	t60 Value (Simon)	t60 value (Slaton)
125	0.74002396	4.47991753	125	1.07139752	3.74147842	125	1.04453223	5.49169914
250	0.57357828	1.1071424	250	0.74391755	1.42245786	250	0.68614801	3.27416644
500	0.62207583	0.94711738	500	0.80411906	0.81534814	500	0.63904086	0.77019764
1000	0.67594484	1.00207448	1000	0.86515595	0.86636719	1000	0.65827991	0.74533645
2000	0.60772781	0.93230849	2000	0.78447281	0.80819736	2000	0.67421595	0.72213598
4000	0.48612726	0.90127325	4000	0.75139892	0.75460745	4000	0.69044016	0.69681657
Average	0.6175796633	1.561638922	Average	0.836743635	1.401409403	Average	0.73210952	1.950058703
Total Average		1.089609293	Total Average		1.119076519	Total Average		1.341084112

Table 3 This shows the experimental data for the study room with the door open with no damping, damping with the 2-ft by 2-ft panels, and damping with the 2-ft by 2-ft panels combined with the 2-ft by 3-ft panels. The T_{60} reverberation time in seconds is given for the octave bands discussed previously along with the average T_{60} value for each laptop and the total average T_{60} time.

Tables 4 and 5 below show the experimental data from the commons area, which we collected using the same methods as in the coaching room. The total average T_{60} time with the door shut and no damping was 2.43-s, but with the door open, it was 1.44-s. Unlike the coaching room, the commons area T_{60} time actually performs better with the coaching room door open as damping tiles are added. The total average T_{60} time using all the damping panels was 1.69-s with the coaching door open and was 1.75-s with the coaching door shut.

Since both spaces need to be optimized, and the coaching room had a much lower reverberation time with the door shut, it made logical sense to try to optimize the commons area with the coaching room door shut. We were able to bring the total average T_{60} time down by 0.68-s by using 176-ft² of damping tiles, but to get the T_{60} time down to the recommended 0.7-s, we would need to bring the reverberation time down by another 1.05-s. If we divide 1.05 by 0.68, we get 1.54, and when we multiply this by the square footage of damping tile used, the result is 272-ft²; this is the square footage of damping tile that needs to be added to the 176-ft² used in the experiment to bring the T_{60} time down to the recommended 0.7-s.

Because of the large amount of space, it would be perfectly feasible to add 448-ft² of damping tiles to the commons area in the student success building to bring the reverberation time down to 0.7-s to meet the ANSI Standard S12.60 recommendation.

Centre of Entry Room - Door Shut - No Damping			Centre of Entry Room - Door Shut - Damping Panels (Small)			Centre of Entry Room - Door Shut - Big and Small Damping Panels			
Octave Bands	t60 Value (Simon)	t60 value (Slaton)	Octave Bands	t60 Value (Simon)	t60 value (Slaton)	Octave Bands	t60 Value (Simon)	t60 value (Slaton) - Mod	t6 value (Slaton) - Raw
125	1.12022675	10.12041364	125	0.85338751	6.77514021	125	1.05390365	7.59017004	8.77445306
250	1.34172621	4.63105944	250	1.23313456	1.44570852	250	1.07060069	1.69813183	3.70772292
500	1.22524612	2.13433208	500	1.27305408	1.56296109	500	1.14826875	1.48792884	3.16622076
1000	1.22146249	1.69757764	1000	1.22226493	1.52820539	1000	1.13033416	1.38529825	2.54687503
2000	1.02042557	2.01471646	2000	1.03592267	1.26371506	2000	1.23984972	1.20383119	3.07972654
4000	0.93091113	1.69748472	4000	0.93553007	1.2132576	4000	0.81668909	1.18615712	3.57577983
Average	1.143333045	3.715930663	Average	1.092215637	2.298164645	Average	1.076607677	2.425252878	
Total Average		2.429631854	Total Average		1.695190141	Total Average		1.750930278	

Table 4. This table shows the experimental data for the common area room with the door shut with no damping, damping with the 2-ft by 2-ft panels, and damping with the 2-ft by 2-ft panels combined with the 2-ft by 3-ft panels. The T_{60} reverberation time in seconds is given for the octave bands discussed previously along with the average T_{60} value for each laptop and the total average T_{60} time.

Centre of Entry Room - Door Open - No Damping			Centre of Entry Room - Door Open - Damping Panels (Small)			Centre of Entry Room - Door Open - Big and Small Damping Panels		
Octave Bands	t60 Value (Simon)	t60 value (Slaton)	Octave Bands	t60 Value (Simon)	t60 value (Slaton)	Octave Bands	t60 Value (Simon)	t60 value (Slaton)
125	0.99009244	3.84632539	125	1.02163774	7.83231122	125	1.01139598	7.33957192
250	1.11684751	1.45179516	250	1.01638279	1.39212874	250	1.16179653	1.340711
500	1.11678707	1.6700041	500	1.04887402	1.494063	500	1.10819584	1.40604104
1000	1.07031791	1.61575247	1000	1.03955555	1.48203833	1000	1.14039864	1.35893183
2000	0.96061526	1.31422539	2000	1.00938802	1.25356081	2000	1.11491662	1.1760881
4000	0.84262369	1.22959651	4000	0.99900141	1.20676372	4000	0.91854292	1.14939887
Average	1.01621398	1.854616503	Average	1.022473255	2.443477637	Average	1.075874422	2.295123793
Total Average		1.435415242	Total Average		1.732975446	Total Average		1.685499108

Table 5 This table shows the experimental data for the common area room with the door open with no damping, damping with the 2-ft by 2-ft panels, and damping with the 2-ft by 2-ft panels combined with the 2-ft by 3-ft panels. The T_{60} reverberation time in seconds is given for the octave bands discussed previously along with the average T_{60} value for each laptop and the total average T_{60} time.

Conclusions:

The physical acoustics of the coaching room and commons area at the UCA Student Success Center were assessed on April 10th, 2024. A series of experiments were conducted to obtain the reverberation time in each room. Then, damping tiles were added to each room to see if the reverberation time could be improved. Based on the data, recommendations were made to improve the speech intelligibility in each space. The recommendations were made based on the ANSI Standard S12.60 for Classroom Acoustics for furnished, unoccupied rooms. For the coaching room 352-ft² of damping material can be added to bring the reverberation time down to the recommended 0.6-s, and for the commons area, 448-ft² of damping material can be added to bring the reverberation time down to the recommended 0.7-s. These recommendations were made based on the assumption that the coaching door in the student success center remains closed. Pictures that show the setup for the experiment and the damping tile placement in each room are included in Appendix A.

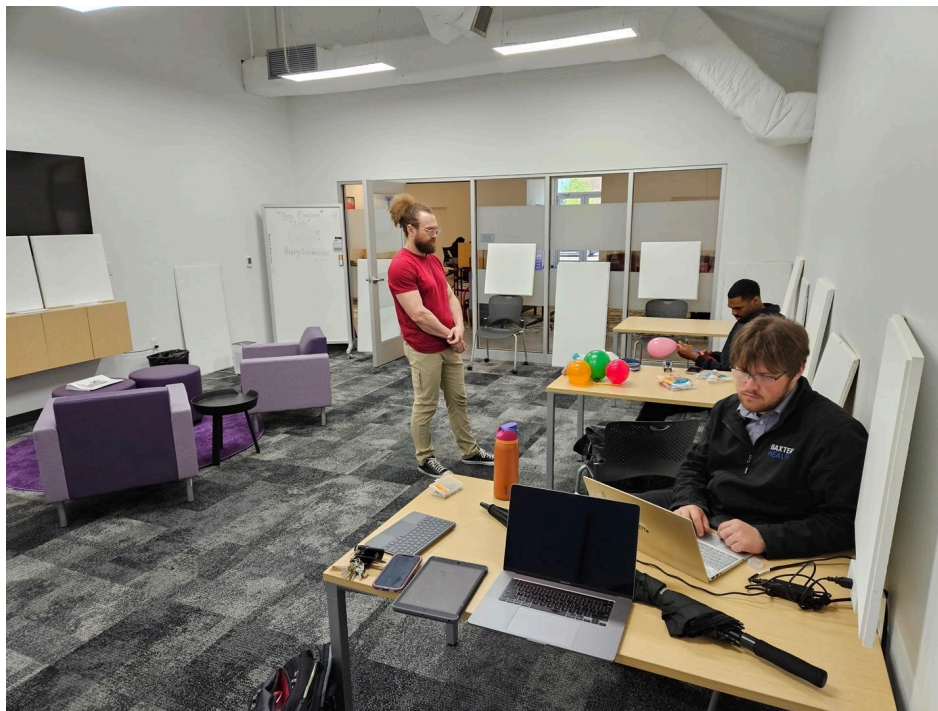
We believe that the physical acoustics of the Student Success Center can be improved by following the recommendations in this report. This will allow students being coached in the Student Success Center to receive an optimal auditory experience.

Appendices:

Appendix A: Photos from the Recording Process

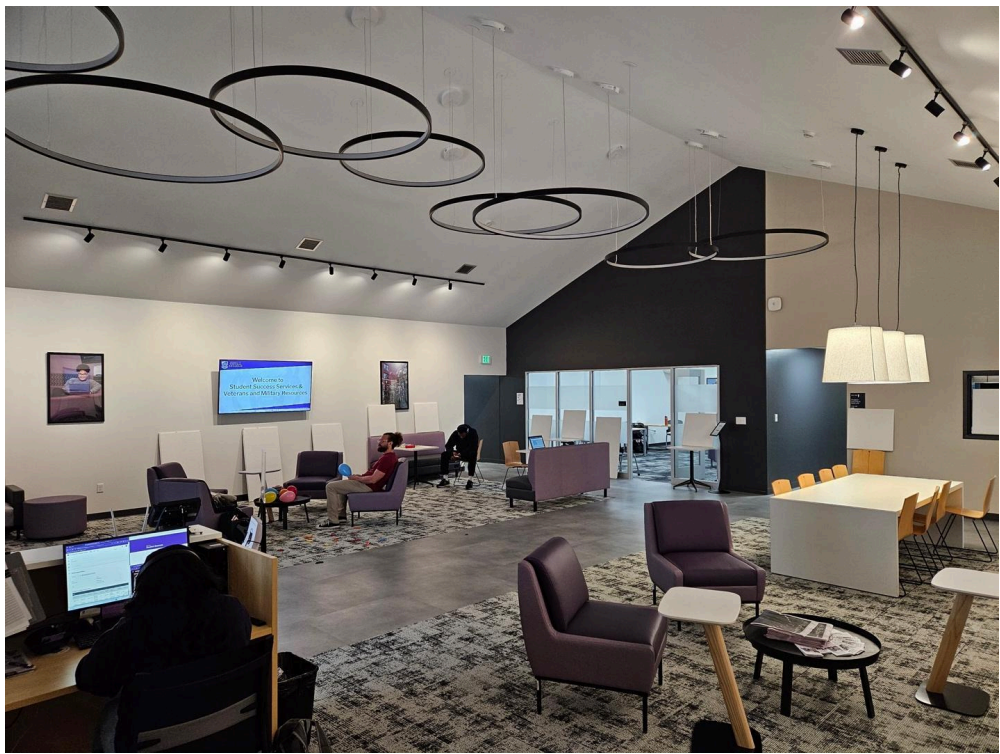












Appendix B: Code:

```
pip install acoustics
```

```
Collecting acoustics
  Downloading acoustics-0.2.6.tar.gz (3.5 MB)
    3.5/3.5 MB 14.5 MB/s eta 0:00:00
    Installing build dependencies ... done
    Getting requirements to build wheel ... done
    Preparing metadata (pyproject.toml) ... done
Requirement already satisfied: numpy>=1.8 in /usr/local/lib/python3.10/dist-packages (from acoustics) (1.25.2)
Requirement already satisfied: scipy>=0.16 in /usr/local/lib/python3.10/dist-packages (from acoustics) (1.11.4)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.10/dist-packages (from acoustics) (3.7.1)
Requirement already satisfied: six>=1.4.1 in /usr/local/lib/python3.10/dist-packages (from acoustics) (1.16.0)
Requirement already satisfied: pandas>=0.15 in /usr/local/lib/python3.10/dist-packages (from acoustics) (2.0.3)
Requirement already satisfied: tabulate in /usr/local/lib/python3.10/dist-packages (from acoustics) (0.9.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.10/dist-packages (from pandas>=0.15->acoustics) (2.8.2)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.10/dist-packages (from pandas>=0.15->acoustics) (2023.4)
Requirement already satisfied: tzdata>=2022.1 in /usr/local/lib/python3.10/dist-packages (from pandas>=0.15->acoustics) (2024.1)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.10/dist-packages (from matplotlib->acoustics) (1.2.1)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.10/dist-packages (from matplotlib->acoustics) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.10/dist-packages (from matplotlib->acoustics) (4.51.0)
Requirement already satisfied: kiwisolver>=1.0.1 in /usr/local/lib/python3.10/dist-packages (from matplotlib->acoustics) (1.4.5)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.10/dist-packages (from matplotlib->acoustics) (24.0)
Requirement already satisfied: pillow>=6.2.0 in /usr/local/lib/python3.10/dist-packages (from matplotlib->acoustics) (9.4.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.10/dist-packages (from matplotlib->acoustics) (3.1.2)
Building wheels for collected packages: acoustics
  Building wheel for acoustics (pyproject.toml) ... done
  Created wheel for acoustics: filename=acoustics-0.2.6-py3-none-any.whl size=68184 sha256=c08d6fc4a266dd4b16ad6cd149a20ef7c59ad01e6a72061016e53b5
  Stored in directory: /root/.cache/pip/wheels/dd/9b/f0/0fa91c83f48277b7ef24800a9930694f78839531b5bab52a42
Successfully built acoustics
Installing collected packages: acoustics
Successfully installed acoustics-0.2.6
```

```
[ ] import numpy as np
    from acoustics import bands,room

    #octave(first octave center frequency,last octave center frequency)

    octave_bands = bands.octave(125,4000)
    print(octave_bands)

[ 125.  250.  500. 1000. 2000. 4000.]

[ ] t60_bands_1 = room.t60_impulse('Big Room - small tiles - coaching door shut - pop 1.wav',octave_bands,rt='t30')
    print(t60_bands_1)

[6.77514021 1.44570852 1.56296109 1.52820539 1.26371506 1.2132576 ]

[ ] t60_bands_2 = room.t60_impulse('Big Room - small tiles - coaching door shut - pop 1.wav',octave_bands,rt='t30')
    print(t60_bands_2)

[6.77514021 1.44570852 1.56296109 1.52820539 1.26371506 1.2132576 ]

▶ t60_bands_3 = room.t60_impulse('Big Room - small tiles - coaching door shut - pop 1.wav',octave_bands,rt='t30')
    print(t60_bands_3)

[6.77514021 1.44570852 1.56296109 1.52820539 1.26371506 1.2132576 ]

▶ t60_average = (t60_bands_1 + t60_bands_2 + t60_bands_3) / 3
    print (t60_average)

[6.77514021 1.44570852 1.56296109 1.52820539 1.26371506 1.2132576 ]
```

References:

- [1] W. Staton, "Mashburn 103 Acoustics ," Google drive: Sign-in, <https://drive.google.com/drive/folders/1hyoVKdKFQH-25WCS-0bq5eeJR-cdVe9s> (accessed Apr. 20, 2024).